

Sakshi Patil

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(Open to Relocate)

SUMMARY

Results-driven **Business Analyst** with **3+ years** of experience in **finance, healthcare, transportation, and e-commerce**. **Proficient in Python, SQL, Tableau, Power BI, Machine Learning, and Statistical Analysis**, with expertise in data-driven decision-making, predictive modeling, and visualization. Built an **AML system for Santander Bank**, cutting manual oversight by **20%**. Developed **12+ healthcare dashboards** and enhanced demand forecasting in the **global seafood trade**. Strong communicator bridging data insights and business strategies, **Actively seeking opportunities** to drive analytics initiatives and enhance data-driven business strategies.

AWARDS, RECOGNITION, CERTIFICATIONS

- **Data Science (ML and AI using Python) – Tech Data Solutions Certification**
- **Deloitte Case Competition 2023- 2nd Place**, Established a Transparent and Equitable Working Capital Fund for a Federal Health Agency's Financial Services
- **EY Info Challenge - 3rd Place**, Analyzed Washington Crash and Seattle/King County Distracted Driver Survey data.

PROFESSIONAL EXPERIENCE

SANTANDER BANK

Business Analyst

Remote, US

Aug 2023 - Dec 2024

Assisting the client in developing an Anti-Money Laundering platform to strengthen compliance with federal and state regulations while improving detection and monitoring capabilities.

- Built an **ETL** pipeline to efficiently extract over **50M+ transaction records** from multiple sources, including **SQL databases** and **CSV** files, ensuring data from diverse systems was integrated into a centralized repository
- Integrated business analysis, **data-driven insights, and solution design to align process** improvements and technology implementation with strategic business objectives
- Utilised stakeholder collaboration, **project management support, and testing expertise** to enhance communication, ensure accurate requirement gathering, and drive successful solution deployment
- Used **statistical analysis** to detect outliers in transaction amounts and assess spending trends, flagging sudden spikes or drops in spending patterns based on recent transaction data
- Wrote **SQL** queries using **CTEs** and **Window Functions** to analyze customer transactions, calculate rolling averages, and identify unusual spending patterns that could indicate fraud
- Implemented **ML algorithms** like **Random Forest, Isolation Forest, and SVM** in **Python** to classify transactions and flag high-risk activities across customer profiles
- Designed an interactive **Power BI** dashboard with **DAX** measures to track fraud risk scores, visualize transaction trends, and provide deeper insights into high-risk customers
- Collaborated with **Model Owners, Risk Management, and Compliance teams** to develop a risk detection model and enhance system functionality, ensuring alignment with **AML/BSA** regulations and addressing regulatory changes
- Educated stakeholders on dashboard usage and model insights, conducting training sessions to help teams interpret risk scores, flagged transactions, and compliance metrics effectively

MARYLAND DEPARTMENT OF TRANSPORTATION

Data Analyst

Maryland, US

Jan 2024 - May 2024

A data integration and analytics project for MTA, consolidating disparate transit datasets to improve trip-level accuracy, operational insights, and real-time decision-making

- Created a unified trip-level view for **MTA** by integrating **1,000,000** rows of trip data from 7 sources, using data modeling techniques to define trips by date, start time, and end time attributes in **MySQL**
- Employed **SSIS** for **ETL** processes to automate the extraction, transformation, and consolidation of the datasets, ensuring consistency and accuracy
- Developed **Power BI** dashboards using **DAX** and **Power Query** for operational insights and optimized **SQL** queries to improve data retrieval efficiency and reporting accuracy

- Coordinated between my team, **UMD faculty, and MDOT stakeholders**, ensuring effective communication and alignment of goals throughout the project for successful execution

DISCOVERU HEALTH

Maryland, US

Data Analyst

May 2023 - Aug 2023

DiscoverU Health faced challenges with low patient engagement, making it difficult to optimize wellness programs, improve retention, and measure the effectiveness of healthcare initiatives.

- Conducted comprehensive market research on three prominent healthcare startups, namely Included Health, One Medical, and Vitable Health
- Queried and analyzed **Salesforce** data with **SQL**, identifying patterns in patient engagement across wellness programs, leading to targeted interventions for improved participation and health outcomes
- Built automated data pipelines in **Azure Data Factory**, integrating claims and clinical data for real-time reporting on program effectiveness and key performance indicators
- Conducted **A/B testing** and chi-square analysis using **R** to evaluate the impact of engagement strategies on patient retention and program effectiveness
- Developed 12 **Power BI** dashboards to track healthcare utilization trends for Medicare Advantage and Medicaid programs, providing insights to improve health outcomes and reduce costs

ARISTON FOOD CONCEPTS

Mumbai, India

Business Intelligence Analyst

Jan 2021 - June 2022

Ariston Food Concept faced challenges in consolidating and analyzing vast amounts of global seafood trade data, leading to inefficiencies in demand forecasting, pricing optimization

- Designed and implemented **ETL processes** on **AWS** to integrate and optimize a global database, streamlining data from buyers, packers, and products
- Conducted **cohort** and **RFM** analyses to evaluate purchasing behavior, delivering insights that refined strategies in the seafood market
- Applied **linear regression** and **time series forecasting**, to enhance demand and pricing forecasts for seafood products, optimize pricing models, and streamline their supply chain operations
- Designed and developed 10+ Power BI dashboards to track seafood trade metrics, automating reporting workflows using **Power BI** and **Excel VBA**, reducing manual reporting time by **30%**

TECH MAHINDRA - MAKERS LAB

Pune, India

Data Scientist

Feb 2020 - Jul 2020

Client: Sweden based Grocery Store

Developed an Automatic Number Plate Recognition system using YOLO V3 and OCR to extract and store vehicle data from live camera feeds for enhanced parking lot security

- Utilized **YOLO V3** and **Python** for real-time localization of number plates from live camera feeds, achieving 97% accuracy with a custom dataset of nearly 20000 annotated images
- Employed advanced image processing techniques to crop and refine number plate images using **Python** and **Pandas**, followed by **OCR** to extract and store alphanumeric data in a structured database

Client: HealthKart - Healthcare and Fitness E-commerce Platform

Data Analyst

HealthKart faced challenges in accurately predicting demand fluctuations and managing inventory efficiently amid the pandemic-driven surge in healthcare e-commerce.

- Centralized data from SQL databases, **Google Analytics**, and transaction logs, processing **10M+** transactions and **50M** product searches during the pandemic surge in online shopping
- Built predictive models using **Random Forest** and **SVM** to forecast demand, reduce stockouts, and minimize churn. This led to a significant increase in customer retention
- Conducted customer segmentation with K-means and Hierarchical Clustering, analyzing 5M+ profiles to identify high-value segments, enabling targeted campaigns, and increasing repeat purchases

PROJECTS

Understanding the Economy of the US using Tableau Prep, Desktop [GitHub](#)

- Analyzed GDP, GNI, inflation, and unemployment using World Bank, US Census, and OECD data to assess COVID-19's impact on the US economy.
- Conducted regional inflation analysis, highlighting disparities between service-driven states (e.g., Maine) and high-income industry states (e.g., Washington).

Impact of Weather, Crashes, & Traffic Violations on Road Safety in Montgomery County Using GCP [GitHub](#)

- Engineered a data pipeline with Python, Cloud Functions, and Pub/Sub to store data from 3 APIs.
- Analyzed data with PySpark on DataProc to evaluate traffic and environmental impacts on collisions.
- Developed predictive models & dashboards with BigQuery, analyzing road safety patterns.

Diabetes Prediction and BMI Prediction Using Machine Learning [GitHub](#)

- Used 100,000+ records from the BRFSS survey to forecast diabetes incidence among participants.
- Utilized various machine learning techniques, such as linear and logistic regression, multivariate analysis, decision trees, SVMs, clustering, and neural networks to develop accurate predictive models.

SKILLS

- **Databases:** Snowflake, Microsoft SQL Server, MySQL
- **Programming Languages:** Python, R, Power Query, DAX
- **Tools & Libraries:** Excel, PowerPoint, JIRA, NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch
- **Data Visualization:** PowerBI, Tableau, Tableau Prep Builder
- **Machine Learning:** Linear and Logistic Regression, Decision Trees, RF, Xgboost, K-means, PCA

EDUCATION

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| • UNIVERSITY OF MARYLAND
<i>Masters in Information Management (GPA: 4/4)</i> | College Park, MD
Aug 2022 - May 2024 |
| • MAHARASHTRA INSTITUTE OF TECHNOLOGY
<i>Btech. Computer Science and Engineering (GPA: 4/4)</i> | Pune, India
Jul 2017 - Jun 2021 |